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EXAMINER
HAM, JIEUN

ART UNIT	PAPER NUMBER
1643	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/253,576	Applicant(s) LAI et al.	
	Examiner Jieun Ham	Art Unit 1643	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 17 November 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,5-6,8-11,13-19,22-27,29 and 31-32 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,5-6,8-11,13-19,22-27,29 and 31-32 is/are rejected.

8) ☑ Claim(s) 14 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 18 May 2023 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Claims 1, 5, 6, 8-11, 13-19, 22-27, 29, and 31-32 are pending in the instant application and being examined on the merit.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

The information disclosure statement filed 7/14/2023 fails to comply with 37 CFR 1.98(a)(2), which requires a legible copy of each cited foreign patent document; each non-patent literature publication or that portion which caused it to be listed; and all other information or that portion which caused it to be listed. It has been placed in the application file, but the information referred to therein has not been considered. The following reference numbers are missing a legible copy: 153-154, 159-160, 162, 164, 176, 178, 191, 213, 215, 220, 261, 288-289, 326, 339, 355, and 445.

Drawings

The drawings are objected to because:

- Instant Figure 11 is a grayscale drawing of Globo H expression in lung cancer (NSCLC) cohort patients, wherein the legend gives some information regarding "positive" and "negative" expression, and based on the figure descriptions, it appears that the different grayscale gradients are supposed to represent the

degree of intensity used in the calculation (page 8, paragraph [0040]). However, the pattern for positive and negative are identical based on the figure legends, and there is no clear indication of what the different gradients represent. Therefore, the figure is not able to be interpreted as currently drawn in grayscale.

- Instant Figure 15 is a grayscale drawing of an overall survival plot (page 9, paragraph [0044]). However, both groups are displayed as the same color, and there are no labels clearly depicting which graph represents which group. Therefore, the figure is not able to be interpreted as currently drawn in grayscale.
- Instant Figure 16 is a grayscale drawing of a progression-free survival plot (page 9, paragraph [0045]). However, both groups are displayed as the same color, and there are no labels clearly depicting which graph represents which group. Therefore, the figure is not able to be interpreted as currently drawn in grayscale.
- Instant Figure 17 is a grayscale drawing of a swimmer plot on EGFR TKIs and OBI-833 duration with various grayscale gradients throughout the whole figure. However, there is no indication of what the different gradients represent. Therefore, the figure is not able to be interpreted as currently drawn in grayscale.
- Instant Figure 18 is a grayscale drawing of tumor response time in NSCLC patients. However, based on the provided legend, it is difficult to identify each patient due to the similarity in pattern and color of the lines. Therefore, the figure is not able to be interpreted as currently drawn in grayscale.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended

replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as “amended.” If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either “Replacement Sheet” or “New Sheet” pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Color photographs and color drawings are not accepted in utility applications unless a petition filed under 37 CFR 1.84(a)(2) is granted. Any such petition must be accompanied by the appropriate fee set forth in 37 CFR 1.17(h), one set of color drawings or color photographs, as appropriate, if submitted via the USPTO patent electronic filing system or three sets of color drawings or color photographs, as appropriate, if not submitted via the via USPTO patent electronic filing system, and, unless already present, an amendment to include the following language as the first paragraph of the brief description of the drawings section of the specification:

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

Color photographs will be accepted if the conditions for accepting color drawings and black and white photographs have been satisfied. See 37 CFR 1.84(b)(2).

Specification

The disclosure is objected to because of the following informalities:

- "Th" should read "The" (page 8, paragraph [0040]).

Appropriate correction is required.

Claim Objections

Claim 14 is objected to because of the following informalities:

- "claims 13" should read "claim 13".
- "rolaxifene" should read "raloxifene".
- "tamoxifen" was written two times. One should be omitted from the list of anti-proliferative agents.
- "letrozole" was written two times. One should be omitted from the list of anti-proliferative agents.
- "bevacizumab" was written two times. One should be omitted from the list of anti-proliferative agents.
- "trastuzumab" was written two times. One should be omitted from the list of anti-proliferative agents.
- "chloranmbucil" should read "chlorambucil".
- "cyclosphosphamide" should read "cyclophosphamide"

Appropriate correction is required.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Claims 1, 5-6, 8-11, 13-17, 26-27, and 31-32 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, because the specification, while being enabling for cancer treatment comprising administering a therapeutically effective dose of Globo H-DT-CRM197 glycoconjugate vaccine, does not reasonably provide enablement for cancer prevention comprising administering a therapeutically effective dose of Globo H-DT-CRM197 glycoconjugate vaccine. The specification does not enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to use the invention commensurate in scope with these claims.

1) Claim 1 is for a method of preventing cancer, but the specification and the state of the art do not teach a method to prevent cancer by administering a Globo H-DT-CRM197 glycoconjugate vaccine. Claims 5-6, 8-11, and 13-17 are further dependent on claim 1 without requiring treatment.
2) Claim 26 is for a method of preventing lung cancer, but the specification and the state of the art do not teach a method to prevent lung cancer by administering a Globo H-DT-CRM197 glycoconjugate vaccine. Claims 27 and 31-32 are further dependent on claim 26 without requiring treatment.

There are many factors to be considered when determining whether there is sufficient evidence to support a determination that a disclosure does not satisfy the enablement requirement and whether any necessary experimentation is "undue." These factors include, but are not limited to:

A. The breadth of the claims;
B. The nature of the invention;
C. The state of the prior art;
D. The level of one of ordinary skill;
E. The level of predictability in the art;
F. The amount of direction provided by the inventor;
G. The existence of working examples; and
H. The quantity of experimentation needed to make or use the invention based on the content of the disclosure.

<u>Scope of the claimed genus and nature of the invention.</u>

1) Claims 1, 5-6, 8-11, and 13-17 are for a method of treating or preventing cancer, e.g. gastric, lung, colorectal, or breast cancer, comprising administering a therapeutically effective dose of Globo H-DT-CRM197 glycoconjugate vaccine.
2) Claims 26-27, and 31-32 are for a method of treating or preventing lung cancer, comprising administering to a patient in need thereof a therapeutically effective dose of Globo H-DT-CRM197 glycoconjugate vaccine by a route selected from:
   a) Administering vaccine two or more times;
   b) Adjusting time interval or dosing amount regimen between two successive administrations;
   c) Adjusting routes of administration or altering injection locations of administration; or
   d) Any combination of the above thereby each administration increases antibody immune response or increases antigen-antibody binding affinity.

State of the Relevant Art; level of one of ordinary skill; and level of predictability of the art.

Cancer has a wide variety of causes, from environmental and/or developmental exposure to ionizing radiation and/or chemical exposure in addition to some types viral and bacterial exposure (Lichtman, 2017, The Oncologist, 22(5): 542–548). Even though cancer is featured by the infinite cell proliferation, its pathogenesis is extremely complex and related to many mechanisms. In general, the hallmarks of cancer consist of ten biological capabilities during the development of cancers, namely sustaining proliferative signaling, evading growth suppressors, resisting cell death, enabling

replicative immortality, inducing angiogenesis, activating invasion and metastasis, tumor-promoting inflammation, genome instability and mutation, evading immune destruction and reprogramming energy metabolism (Hanahan et al, 2011, Cell, 144(5): 646-674, Figures 1 and 3). More than 100 types of cancer have been identified which are typically termed for the organs or tissues where they occur. Hanahan taught over the past decade, tumors have increasingly been recognized as organs whose complexity approaches and may even exceed that of normal healthy tissues (page 661, right column second to last paragraph). Hanahan taught many human tumors are histopathologically diverse, containing regions demarcated by various degrees of differentiation, proliferation, vascularity, inflammation, and/or invasiveness (page 662, left column, first paragraph).

Clinical trials aimed at proving preventative cancer activity attributable to a specific intervention are largely infeasible, due to the impossibly large number of subjects and an equally impossible long timeframe. Although cancer is a common disease, specific types of cancer are still relatively infrequent events in an otherwise healthy population. Therefore, trials with cancer incidence as endpoints would necessarily involve several thousands of subjects followed for several decades. Such logistic difficulties have precluded cancer prevention trials with cancer incidence as an endpoint in all but a selected few malignancies for treatments such as tamoxifen and finasteride (Lee et al, 2011, Nature Reviews Cancer, 11: 211-218; page 211, left column last paragraph) Lee further taught although many reports have suggested benefits and targets of phytochemicals, these reports mainly rely on cell and animal models (page 216, right column last paragraph). Lee taught that in order to apply phytochemicals as

personalized cancer preventive agents, the effects of phytochemicals in humans will need to be assessed (page 216, right column last paragraph).

The development of anticancer vaccines requires the identification of unique epitope markers, preferably expressed exclusively on the surface of cancer cells. The development of a vaccine conjugate and adjuvant design targeting Globo H started with the development of a carbohydrate-based vaccine for metastatic breast cancer, leading to the discovery that the Globo H-targeted vaccine additionally targeted SSEA3 and SSEA4, and these three glycolipids were uniquely expressed on the cell surface of 15 additional cancer types in addition to breast cancer (Danishefsky et al, 2015, Accounts of Chemical Research, 48(3): 643-652; page 643, Abstract). Danishefsky taught that in preclinical and clinical studies, the Globo H vaccine generated high-titer IgM and modest IgG responses in mice wherein these antibodies were selectively reactive against Globo-H positive cell lines and were observed to induce complement-mediated lysis of Globo H-positive MVF-7 cells (page 647, last paragraph – 648, first paragraph). On the basis of these encouraging data, a clinical trial was also carried out using the Globo H vaccine in patients with metastatic breast cancer, resulting in the generation of IgM antibodies in most patients along with a relatively weak IgG response with evidence of complement-dependent cytotoxicity (CTC) ins several patients (page 648, second paragraph – page 649, first paragraph). Danishefsky further teaches when Globo-H was conjugated to the diphtheria toxin mutant CRM197 and used together with a glycolipid adjuvant, the vaccine gave the best immune response (Figure 10; Figure 11; and page 650, first paragraph). However, the functional mechanism of the Globo-series glycosphingolipids associated with tumor metastasis is still not well understood.

Furthermore, Danishefsky teaches that the mechanism of action of the Globo-H vaccine, including its uptake, processing, and presentation in the immune system, remains to be investigated in order to elucidate the process of immune response and cross reactivity toward the unique glycan epitopes (page 650, last paragraph – 651, first paragraph).

Summary of Species disclosed in the original specification; the amount of direction provided by the inventor, existence of working examples; and quality of experimentation needed to make or use the invention based on the content of the disclosure.

1) The specification only identifies cancer treatment in Figures 10-18, showing immune responses to the vaccine (Figures 10, 12-14), Globo H expression in lung cancer NSCLC cohort patients (Figure 11), overall survival and progression free survival (Figures 15 and 16, respectively) in OBI-833 dose escalation phase and cohort expansion phase, swimmer plot on EGFR TKIs and OBI-833 treatment duration in Figure 17, and tumor response over time of NSCLC patients (Figure 18). The treated groups already had cancer before treatment. There are no cancer prevention studies in the instant specification.
2) The instant specification taught: a) pharmacology studies that showed active immunotherapy with OBI-833/OBI-821 can effectively stimulate anti-Globo H IgM and IgG responses in an in vivo murine model. Mouse vaccinated with OBI-833/OBI-821 can significantly inhibit tumor growth rate in an in vivo subcutaneous Globo H positive tumor implant model. It was also observed that the treatment of standard chemotherapeutic agent, gemcitabine, followed by vaccinated with OBI-833/OBI-

821, did not affect the production of anti-Globo H IgM and IgG antibodies induced by OBI-833/OBI-821 (page 30-31, paragraph [0111]; and b) no subject achieved the best tumor response as complete response or partial response in this study (page 100, paragraph [0379]).

3) The instant specification concludes that the results of this study could demonstrate the safety, immune response, and preliminary clinical efficacy of OBI-833/OBI-821 in NSCLC patients (page 100, paragraph [0379]).

Conclusion

1) The Applicant does not have enablement for a method of preventing cancer, e.g. gastric, lung, colorectal, or breast cancer, comprising administering a therapeutically effective dose of Globo H-DT-CRM197 glycoconjugate vaccine in claims 1, 5-6, 8-11, and 13-17.
2) The Applicant does not have enablement for a method of preventing lung cancer, comprising administering to a patient in need thereof a therapeutically effective dose of Globo H-DT-CRM197 glycoconjugate vaccine by a route selected from:
   a) Administering vaccine two or more times;
   b) Adjusting time interval or dosing amount regimen between two successive administrations;
   c) Adjusting routes of administration or altering injection locations of administration; or
   d) Any combination of the above thereby each administration increases antibody immune response or increases antigen-antibody binding affinity;

in claims 26-27, and 31-32.

> The following is a quotation of 35 U.S.C. 112(b):
> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

> The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 14, 26 and 29 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 14 contains the trademark/trade name Camptothecin, Sutent, Cremophor, Novantrone, Angiozyme, Allovectin, Vaxid. Where a trademark or trade name is used in a claim as a limitation to identify or describe a particular material or product, the claim does not comply with the requirements of 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph. See *Ex parte Simpson*, 218 USPQ 1020 (Bd. App. 1982). The claim scope is uncertain since the trademark or trade name cannot be used properly to identify any particular material or product. A trademark or trade name is used to identify a source of goods, and not the goods themselves. Thus, a trademark or trade name does not identify or describe the goods associated with the trademark or trade name. In the present case, the trademark/trade name is used to identify/describe an anti-proliferative agent wherein the anti-proliferative agent is administered to the patient in

combination with a Globo H-DT-CRM197 conjugate vaccine and, accordingly, the identification/description is indefinite.

Regarding instant claim 26, the phrase "for example" renders the claim indefinite because it is unclear whether the limitation(s) following the phrase are part of the claimed invention. See MPEP § 2173.05(d).

Instant claim 29 recites the limitation "The method of claim 28" in the preamble of claim 29. There is insufficient antecedent basis for this limitation in the claim, since claim 28 is a canceled claim. Therefore, claim 29 is indefinite as being incomplete by its dependence on cancelled claim 28. See MPEP 608.01(n).

The following is a quotation of 35 U.S.C. 112(d):

> (d) REFERENCE IN DEPENDENT FORMS.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

The following is a quotation of pre-AIA 35 U.S.C. 112, fourth paragraph:

> Subject to the following paragraph [i.e., the fifth paragraph of pre-AIA 35 U.S.C. 112], a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

Claim 29 is rejected under 35 U.S.C. 112(d) or pre-AIA 35 U.S.C. 112, 4th paragraph, as being of improper dependent form for failing to further limit the subject matter of the claim upon which it depends, or for failing to include all the limitations of the claim upon which it depends. Because claim 29 depends from a cancelled claim, the claim fails to incorporate by reference all the limitations of the claim to which it

refers. Similarly, because of the dependency upon cancelled claim 28, claim 29 fails to specify further limitation. Applicant may cancel the claim(s), amend the claim(s) to place the claim(s) in proper dependent form, rewrite the claim(s) in independent form, or present a sufficient showing that the dependent claim(s) complies with the statutory requirements.

For the purpose of expedited prosecution, claim 29 is interpreted to depend from claim 26.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claims 1,5, 9-11, 15, 18-19, 23, 25-27, and 31 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Wong et al (WO2010/005598 A1, IDS entered on 1/18/2024; hereinafter Wong). Wong et al teaches the following:

Regarding instant claim 1, Wong teaches a method of treating cancer (e.g. breast cancer or lung cancer) by administering to a subject in need thereof an effective amount of an immune composition (or vaccine) (page 8, paragraph [0021]; pages 24-25, paragraph [0107]; page 9, paragraph [0027]), wherein the vaccine is a glycoconjugate vaccine comprising the B cell epitope, Globo H, which is conjugated to the immunogenic carrier diphtheria toxin cross-reacting material, DT-CRM197 (page 3, paragraph [002]; page 28, paragraph [0115]; claims 1 and 2).

Regarding instant claims 5 and 9, Wong teaches the immune composition comprising Globo H conjugated to the protein carrier, DT-CRM197 (hereinafter GH-CRM197) is mixed with a pharmaceutically acceptable carrier to form an immune composition (e.g. vaccine) via conventional methods, wherein the immune composition herein can be administered parenterally, e.g. intravenous injection, subcutaneous injection or intermuscular injection (page 28, paragraph [115]).

Regarding instant claims 10 and 11, Wong teaches that groups of mice were immunized intramuscularly three times at two weeks interval (i.e. once every two weeks, three times) with the GH-CRM197 vaccine to perform serologic assays (page 47, paragraph [0180]).

Regarding instant claim 15, Wong teaches a method of treating cancer by administering to a subject in need thereof an effective amount of the GH-CRM197 vaccine, wherein the treatment with an effective amount of the cancer vaccine inhibits

tumor growth or reduces the size of a tumor (page 8, paragraph [0023]; page 9, paragraph [0027]).

Regarding instant claims 18, 19, 23, and 25, Wong teaches an immunogenic composition comprising a glycan consisting essentially of Globo H wherein the glycan is conjugated with the carrier protein, DT-CRM197, via a linker, wherein the immunogenic composition induces an immune response that induces higher relative levels of IgG antibodies as compared to IgM isotype antibodies in a subject (pages 6-7, paragraphs [0013]-[0014]; Figures 7 and 12A) wherein the subject is human (page 20, paragraph [0092]; page 23, paragraph [101]).

Regarding instant claims 26 and 27, Wong teaches a method of treating lung cancer (e.g. small cell lung cancer) by administering to a subject in need thereof an effective amount of an immune composition wherein the immune composition is GH-CRM197 (pages 24-27, paragraphs [0107]-[0109]; pages 28-29, paragraph [0115]) wherein the immune composition comprises at least one adjuvant to induce a humoral or cellular immune response (page 7, paragraph [0016]), and the immune composition is administered three times with weekly intervals (page 47, paragraph [0180]).

Regarding instant claim 29, Wong teaches a method of treating lung cancer (e.g. small cell lung cancer) by administering to a subject in need thereof an effective amount of an immune composition wherein the immune composition is GH-CRM197 (pages 24-27, paragraphs [0107]-[0109]; pages 28-29, paragraph [0115]) wherein the vaccine is administered three times with weekly intervals (page 47, paragraph [0180]) and the immune response comprises an IgM and IgG response (page 41, paragraph [0160]).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claims 6, 22, and 32 are rejected under 35 U.S.C. 103 as being unpatentable over Wong et al (WO2010/005598 A1, IDS entered on 1/18/2024; hereinafter Wong) as applied to claim 1 above, and further in view of Yu et al (WO2019/217951 A1; hereinafter Yu).

The teachings of Wong are discussed in the 102 rejection above.

However, Wong does not teach a method of treating cancer (e.g. breast cancer or lung cancer) by administering to a subject in need thereof an effective amount of a pharmaceutical composition comprising an OBI-833 vaccine and an OBI-821 adjuvant (instant claim 6). Additionally, Wong does not teach a method for inducing an immune response in a subject, comprising administering to the subject an immunogenic agent wherein the immunogenic agent is OBI-833 and related variants (instant claim 22). Furthermore, Wong does not teach a method of treating lung cancer (e.g. small cell lung cancer) by administering to a subject in need thereof an effective amount of a vaccine wherein the vaccine is OBI-833 and further comprises OBI-821 adjuvant (instant claim 32).

The deficiency is resolved by Yu et al.

Yu teaches a therapeutic cancer vaccine, OBI-833/821, wherein OBI-833, a Globo H-DT glycoconjugate, is combined with adjuvant OBI-821, a saponin adjuvant (page 11, paragraphs [0032] and [0034]). Yu also teaches that when patients were immunized with OBI-833/821 in a phase 1 clinical study (Trial Identifier #NCT02310464), various levels of total IgM and/or Globo H IgM were induced (pages 15-16, paragraphs [0051]-[0058]; Tables 1 and 2; Figures 4A and 4B). Furthermore, Yu teaches that the cancer vaccine OBI-833/821 was administered for treating Globo

series antigen expressing cancer, e.g. lung cancer (pages 10-11, paragraphs [0031]-[0032]).

Regarding instant claim 6, it would have been obvious for a person having ordinary skill in the art at the time of filing to modify the method of treating cancer (e.g. breast cancer or lung cancer) by administering to a subject in need thereof an effective amount of an immune composition (or vaccine) wherein the vaccine is a glycoconjugate vaccine comprising a B cell epitope, Globo H conjugated to an immunogenic carrier diphtheria toxin cross-reacting material, DT-CRM197 as taught by Wong and substitute the immune composition with the OBI-833 vaccine and a OBI-821 adjuvant as taught by Yu. This is obvious because, Wong teaches a method of treating cancer, e.g. breast or lung cancer, by administering to a subject in need thereof an effective amount of an immune composition comprising a vaccine wherein the vaccine comprises of a B cell epitope, Globo H, conjugated to an immunogenic carrier DT-CRM197, and Yu teaches a therapeutic cancer vaccine, OBI-833/821, wherein OBI-833 is a Globo H-DT glycoconjugate and OBI-821 is a saponin adjuvant. Therefore, it is obvious to a skilled artisan with reasonable expectation of success to have been motivated to take the method of treating cancer (e.g. breast or lung cancer) by administering to a subject in need thereof an effective amount of an immune composition as taught by Wong wherein the immune composition comprises the vaccine OBI-833 and an OBI-821 adjuvant as taught by Yu to form the instant method for treating advanced lung or breast cancer in a patient comprising administering a therapeutically effective dose of a pharmaceutical composition comprising an OBI-833 vaccine and an OBI-821 adjuvant.

Regarding instant claim 22, it would have been obvious for a person having ordinary skill in the art at the time of filing to modify the method of inducing an immune response in a subject comprising administering to the subject an immunogenic agent as taught by Wong wherein the immunogenic agent is OBI-833/821 as taught by Yu. This is obvious because, Wong teaches a method comprising an immunogenic composition wherein the immunogenic composition induces an immune response that induces higher relative levels of IgG antibodies as compared to IgM isotype antibodies in a subject, and Yu teaches a therapeutic cancer vaccine, OBI-833/821 wherein patients immunized with OBI-833/821 had induced levels of total IgM or Globo H IgM. Therefore, it is obvious to a skilled artisan with reasonable expectation of success to have been motivated to take the method of inducing an immune response in a subject comprising administering to the subject an immunogenic agent as taught by Wong wherein the immunogenic agent is OBI-833/821 as taught by Yu to form the instant method for inducing an immune response in a subject, comprising administering to the subject an immunogenic agent wherein the instant immunogenic agent comprises OBI-833 and related variants.

Regarding instant claim 32, it would have been obvious for a person having ordinary skill in the art at the time of filing to modify the method of treating lung cancer (e.g. small cell lung cancer) by administering to a subject in need thereof an effective amount of an immune composition wherein the immune composition (or vaccine) is GH-CRM197 wherein the vaccine is administered three times with weekly intervals as taught by Wong and substitute the immune composition with OBI-833 and an OBI-821 adjuvant as taught by Yu. This is obvious because, Wong teaches a method of treating

lung cancer (e.g. small cell lung cancer) by administering to a subject in need thereof an effective amount of an immune composition wherein the immune composition is GH-CRM197 wherein the vaccine is administered three times with weekly intervals, and Yu teaches that the cancer vaccine OBI-833/821 was administered to a subject in need thereof for treating Globo series antigen expressing cancer, e.g. lung cancer. Therefore, it is obvious to a skilled artisan with reasonable expectation of success to have been motivated to take the method of treating lung cancer (e.g. small cell lung cancer) by administering to a subject in need thereof an effective amount of an immune composition as taught by Wong wherein the immune composition is OBI-833 and an OBI-821 adjuvant as taught by Yu to form the instant method for treating lung cancer comprising administering to a patient in need thereof a therapeutically effective dose of a vaccine by a route selected from administering vaccine two or more times, wherein the instant vaccine comprises OBI-833 and further comprises OBI-821 adjuvant.

Claims 8, 13-14, and 16-17 are rejected under 35 U.S.C. 103 as being unpatentable over Wong et al (WO2010/005598 A1, IDS entered on 1/18/2024; hereinafter Wong) as applied to claim 1 above, and further in view of Yu et al (WO2018/022933 A1, IDS entered on 7/14/2023; hereinafter Yu '933) and Wu (WO2016/118191 A1; hereinafter Wu).

The teachings of Wong are discussed in the 102 rejection above.

However, Wong does not teach a method for treating cancer in a subject comprising administering a therapeutically effective dose of a Globo H-DT-RM197 glycoconjugate vaccine, wherein the therapeutically effective dose is less than 1000μg

(instant claim 8) and the vaccine is administered to the patient in combination with one or more anti-proliferative agent, e.g. bevacizumab (instant claims 13 and 14). Furthermore, Wong does not teach the method above further comprising reducing tumor volume or improving survival of the subject by modulating Globo series antigens interaction, wherein the survival comprises overall survival (OS) or progression-free survival (PFS) (instant claim 16), and the modulation of Globo series antigens interaction further comprises an induction of antibody-dependent cellular cytotoxicity (ADCC) and complement-dependent cytotoxicity (CDC) for tumor killing (instant claim 17).

The deficiency is resolved by Yu '933 et al.

Yu '933 teaches a method of treating cancer in a patient in need thereof comprising administering to the patient a therapeutically effective amount of a therapeutic composition comprising a Globo series antigen, e.g. Globo H, conjugated to a carrier moiety, e.g. DT-CRM197 via a linker (page 38, paragraph [0113]; page 4, paragraphs [005]-[006]), wherein the therapeutically effective dose is 0.1μg/kg of patient body weight (pages 37-38, paragraphs [0111]-[0112]) and an anti-proliferative drug, e.g. bevacizumab, is co-administered in combination with the therapeutic composition (page 24, paragraph [076]; page 9, paragraph [019]-[020]). Furthermore, Yu '933 teaches the method mentioned above further comprising reducing tumor volume or improving the survival of a subject, wherein the survival comprises overall survival or progression free survival over a control placebo (page 38, paragraph [113]).

Wu teaches an immunogenic composition capable of eliciting an immune response against a cancer cell (e.g. lung cancer or breast cancer cell) in a subject,

wherein the immune response includes generation of antibodies that specifically bind to one or more of the antigens selected from the group consisting of Globo H, SSEA3, and SSEA4, wherein the antibodies are developed to target one or more of Globo H, SSEA3, and SSEA4 expressed on the surface of cancer cells to trigger CDC or ADCC to kill these cells (pages 10-11, paragraphs [0032]-[0033]).

Regarding instant claims 8, 13, and 14, it would have been obvious for a person having ordinary skill in the art at the time of filing to modify the method of treating cancer (e.g. breast cancer or lung cancer) by administering to a subject in need thereof an effective amount of an immune composition (or vaccine) wherein the vaccine is a glycoconjugate vaccine comprising a B cell epitope, Globo H, conjugated to an immunogenic carrier diphtheria toxin cross-reacting material, DT-CRM197 as taught by Wong, to include co-administration of an anti-proliferative drug, e.g. bevacizumab, in combination with the therapeutic composition wherein the therapeutically effective dose as 0.1μg/kg of patient body weight as taught by Yu '933. This is obvious because, Wong teaches a method of treating cancer, e.g. breast or lung cancer, by administering to a subject in need thereof an effective amount of an immune composition comprising a vaccine wherein the vaccine comprises of a B cell epitope, Globo H, conjugated to an immunogenic carrier DT-CRM197, and Yu '933 teaches a method of treating cancer in a patient in need thereof comprising administering to the patient a therapeutically effective amount of a therapeutic composition comprising a Globo series antigen, e.g. Globo H, conjugated to a carrier moiety, e.g. DT-CRM197 via a linker, wherein the therapeutically effective dose is 0.1μg/kg of patient body weight and an anti-proliferative drug, e.g. bevacizumab, is co-administered in combination with the therapeutic composition.

Therefore, it is obvious to a skilled artisan with reasonable expectation of success to have been motivated to take the method of treating cancer, e.g. breast or lung cancer, by administering to a subject in need thereof an effective amount of an immune composition comprising a vaccine wherein the vaccine comprises of a B cell epitope, Globo H, conjugated to an immunogenic carrier DT-CRM197 as taught by Wong, wherein the therapeutically effective dose is 0.1μg/kg of patient body weight, and an anti-proliferative drug, e.g. bevacizumab, is co-administered in combination with the therapeutic composition as taught by Yu '933 to form the instant method of treating advanced lung or breast cancer in a patient comprising administering a therapeutically effective dose of a pharmaceutical composition comprising a vaccine wherein the vaccine comprises Globo H conjugated to DT-CRM197 wherein the instant therapeutically effective dose is less than 1000μg and the instant vaccine is administered to the patient in combination with an anti-proliferative agent, bevacizumab.

Regarding instant claims 16 and 17, it would have been obvious for a person having ordinary skill in the art at the time of filing to modify the method of treating cancer (e.g. breast cancer or lung cancer) by administering to a subject in need thereof an effective amount of an immune composition comprising a vaccine, wherein the vaccine comprises Globo H conjugated to DT-CRM197 as taught by Wong, to further comprise reducing tumor volume and improving survival of the subject, wherein the survival comprises overall survival or progression-free survival as taught by Yu '933, by generating an immune response comprising modulating Globo series antigen interaction by generating antibodies that specifically bind to Globo H, wherein the antibodies are developed to target Globo H expressed on the surface of cancer cells to trigger CDC or

ADCC to kill these cells as taught by Wu. This is obvious because, Wong teaches a method of treating cancer, e.g. breast or lung cancer, by administering to a subject in need thereof an effective amount of an immune composition comprising a vaccine wherein the vaccine comprises of a B cell epitope, Globo H, conjugated to an immunogenic carrier DT-CRM197, Yu '933 teaches a method of treating cancer in a patient in need thereof comprising administering to the patient a therapeutically effective amount of a therapeutic composition comprising a Globo series antigen, e.g. Globo H, conjugated to a carrier moiety, e.g. DT-CRM197, and reducing tumor volume or improving survival of a subject, wherein the survival comprises overall survival or progression free survival over a control placebo, and Wu teaches an immunogenic composition capable of eliciting an immune response against a cancer cell (e.g. lung cancer or breast cancer cell) in a subject, wherein the immune response includes generation of antibodies that specifically bind to Globo H, wherein the antibodies are developed to target Globo H expressed on the surface of cancer cells to trigger CDC or ADCC to kill these cells. Therefore, it is obvious to a skilled artisan with reasonable expectation of success to have been motivated to take the method of treating cancer, e.g. breast or lung cancer, by administering to a subject in need thereof an effective amount of an immune composition comprising a vaccine wherein the vaccine comprises Globo H conjugated to DT-CRM197 as taught by Wong and further comprise reducing tumor volume and improving survival of the subject, wherein the survival comprises overall survival or progression-free survival as taught by Yu '933, by generating an immune response comprising modulating Globo series antigen interaction by generating antibodies that specifically bind to Globo H expressed on the surface of cancer cells to

trigger CDC or ADCC as taught by Wu to form the instant method of treating advanced lung or breast cancer in a patient comprising administering a therapeutically effective dose of a pharmaceutical composition comprising a vaccine wherein the vaccine comprises Globo H conjugated to DT-CRM197 wherein the instant method further comprises reducing tumor volume and improving survival of the subject by modulating Globo series antigens interaction, wherein the survival comprises overall survival or progression-free survival and the modulation of Globo series antigens interaction further comprises generating an immune response comprising modulating Globo series antigen interaction by generating antibodies that specifically bind to Globo H expressed on the surface of cancer cells to trigger CDC or ADCC.

Conclusion

No claims are allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Jieun Ham whose telephone number is (571)272-7779. The examiner can normally be reached Monday - Thursday 7-3.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Julie Wu can be reached at (571) 272-5205. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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